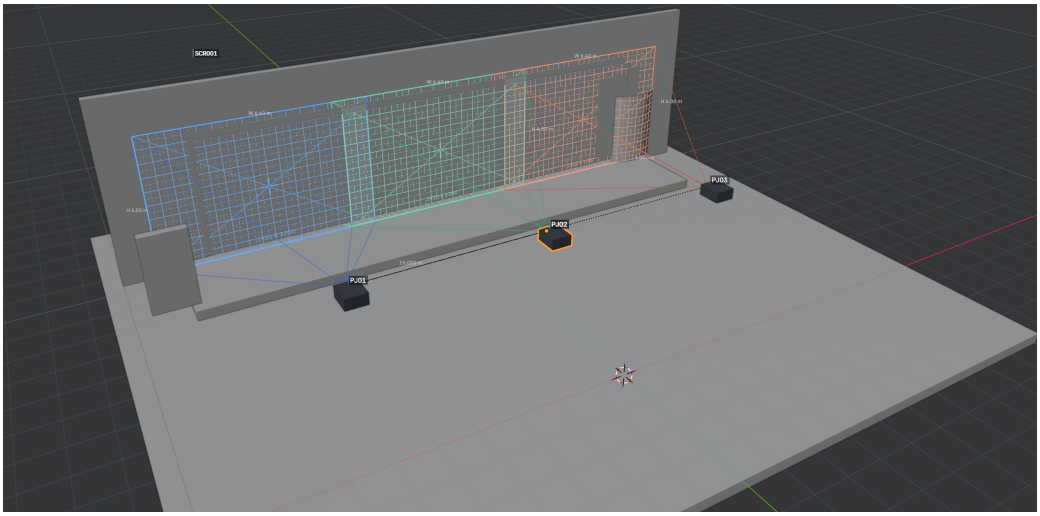
Beam Training Project Guide

Start with a working copy

Open Beam_Training_Project.blend with the guide's Beam build enabled. File → Save As to a new name. Use metres. The original file resets the exercises; analysis caches are recalculated after opening.



1 Find every module

Locate Projector, Screen, Blend, Views and Export in the inspector and left rail. Press T over the 3D View if the rail is hidden. Expand BEAM collections in the Outliner.

2 Select PJ02

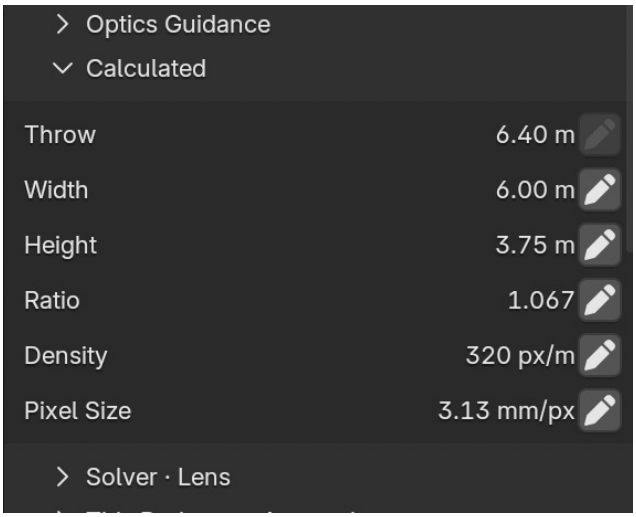
Select PJ02 from BEAM → Projectors. Confirm Active: PJ02 in Projector. It is a Generic / Manual example; PJ01 uses the existing Panasonic PT-RZ990 / ET-DLE105 preset.

3 Move PJ02

With pointer in the 3D View press G, X, 1, Enter. Observe the moved image. Undo with Command-Z on Mac (Ctrl-Z on other Blender setups) before continuing. Check that PJ02 returned.

Change the image and solve

The starting raster is 1920 × 1200. PJ02 starts at approximately 6.4 m throw with ratio 1.00 and image width 6.4 m.



4 Change throw ratio

In Projection set Throw Ratio to 1.20. Confirm Width about 5.33 m, Height about 3.33 m and Density about 360 px/m.

5 Use the inverse solver

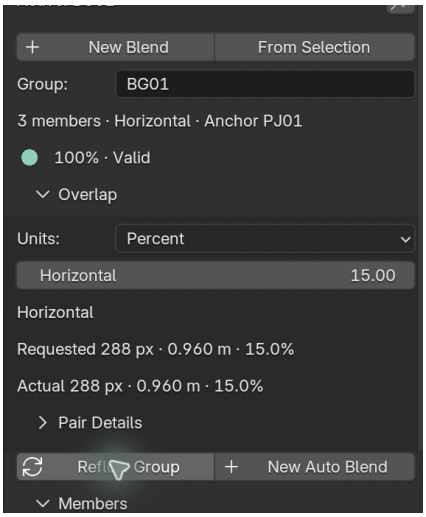
Click the Width pencil and enter 6 m. Solve By: Lens / Throw Ratio. Keep Target locked and Lens unlocked. Apply with the checkmark. Confirm Width 6 m, Height 3.75 m, Throw 6.40 m and Ratio about 1.067. Restore Throw Ratio to 1.00 after this exercise.

6 Adjust BG01 overlap

Select BG01 or open its Blend context. Change requested horizontal overlap from 12.5% to 15%. This is the desired join; inspect the result after reflow.

Reflow and measure coverage

Keep the changed 15% overlap for the following comparison. Use the same receiver, source set and sample count for both measurements.



7 Reflow BG01

Click Reflow Group. Confirm PJ01, PJ02 and PJ03 remain members. Requested and actual horizontal overlap should read 288 px, 0.960 m and 15%.

8 Run Density analysis

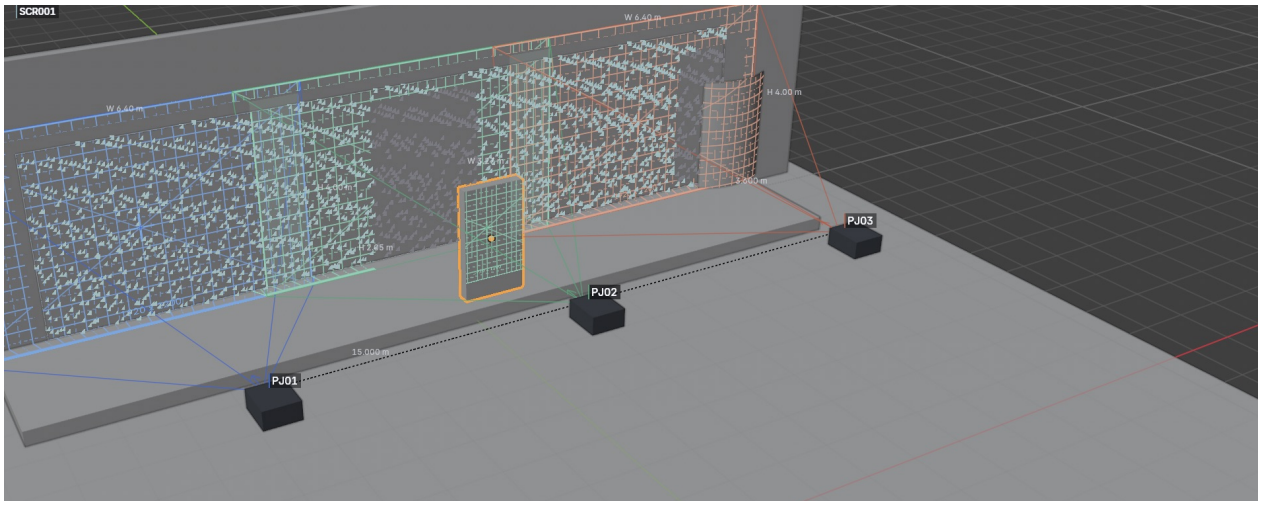
Expand Surface Analysis. Set Receiver SCR001, Projectors Active Blend Group and Samples Standard · 2048. Click Analyse Surface. Set Heatmap to Density. Confirm sources PJ01, PJ02, PJ03; visible coverage is about 93.1% with the blocker at X = -9 m. Visible samples are about 300 px/m.

9 Move the blocker

Select TRAINING_Occluder in the Outliner. In the 3D View press G, X, 9, Enter. It moves from X = -9 to X = 0 m. Leave it there for the next exercises.

Compare occlusion and stretch

The geometry has changed. The old analysis is stale and must be refreshed.



10 Reanalyse and compare

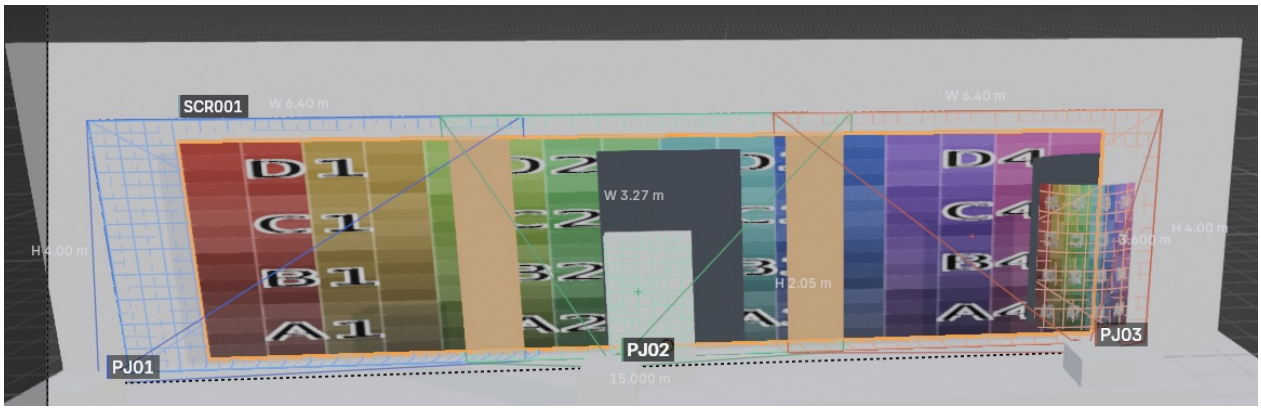
With SCR001 and the same group/settings, click Analyse Surface again. Visible coverage falls to about 78.6%; occluded samples rise to 21.4%. Nominal coverage remains 100%. A blocker at X = -2 lies in an overlap region and can leave total coverage unchanged; use X = 0 for this demonstration.

11 Inspect Stretch

Choose Stretch in Heatmap. The flat screen is close to 1:1. Then set Receiver to TRAINING_Curved_Receiver and Analyse Surface. Recall the Stretch view for a close angle. Inspect the changing surface angle and overlay; the tested curved result has 100% sampled coverage and roughly 214–384 px/m density. Turn Show Heatmap off before reviewing the receiver texture.

Review mapped content

The colour chart is packed into the .blend and the receiver has valid UVs. Use the receiver’s material in Material Preview.



12 Enable Reverse Mapping

Recall the Reverse Mapping study view. Select SCR001 in the Outliner. Expand Reverse Mapping, confirm Receiver SCR001, choose Coverage From: Blend Group and BG01. Click Enable on Receiver. Use the Material Preview sphere at the upper right of the 3D View.

13 Try all four modes

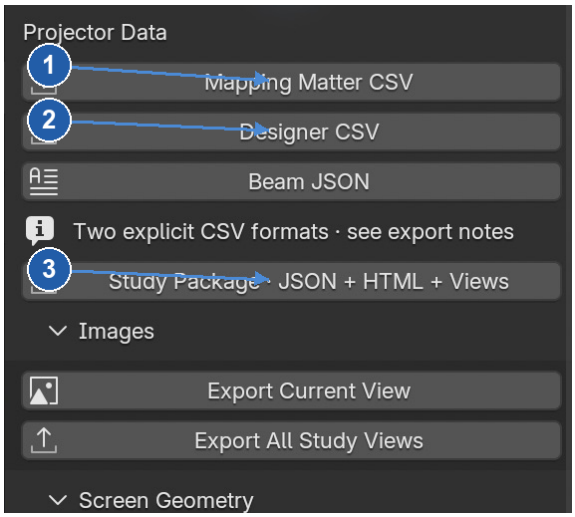
Compare Covered Only, Uncovered Dimmed, Uncovered Hidden and Overlap Diagnostic. With the blocker at X = 0, the central missed region is visible. Dimmed retains content at Outside Brightness; Hidden reveals geometry behind it; Overlap Diagnostic marks joins amber. Finish with Restore Materials.

14 Recall Study Views

Use View beside Overview, Front, Blend, Reverse Mapping and Density; scroll the list to Stretch. Use Exit to leave camera view. View recall does not change the analysis receiver or rebuild its result.

Export the training study

Use distinct filenames and keep sidecars with their exports. These steps complete the Beam side of the handoff.



15 Export a Density view

Set Surface Analysis Receiver back to SCR001, Active Blend Group, Standard · 2048 and Analyse Surface. Recall Density. Confirm This View · Export Annotations → Analysis Overlay: Density. Export This View. Check the PNG and .analysis.json sidecar; the latter records receiver, sources and statistics.

16 Export Mapping Matter CSV

Open Export → Mapping Matter CSV. Save Beam_projectors.csv. Check three PJ rows and keep Beam_projectors.metadata.json.

17 Export Designer CSV

Use the separate Designer CSV button. Save Beam_Designer.csv and keep its metadata sidecar. Its columns differ from Mapping Matter; do not substitute one for the other.

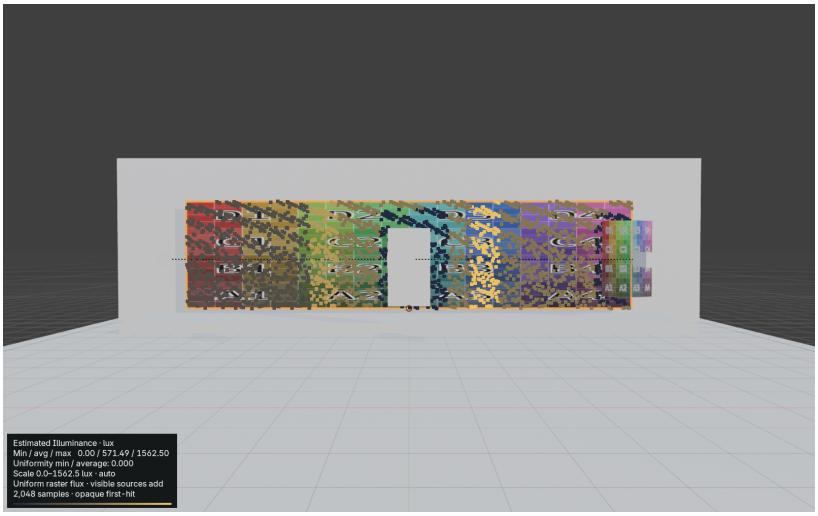
18 Export a Study Package

Choose Study Package · JSON + HTML + Views. Enter a new folder name, leave Render Study Views on and export. Check beam-study.json, index.html and nine view images. The JSON includes the technical study; PATCH document composition is a separate integration still in testing.

Save your exercise copy. Reopen the original training file to reset projector placement, overlap and blocker position.

Estimate light and compare a shadow

Use the same receiver, sources and sample count for both runs.

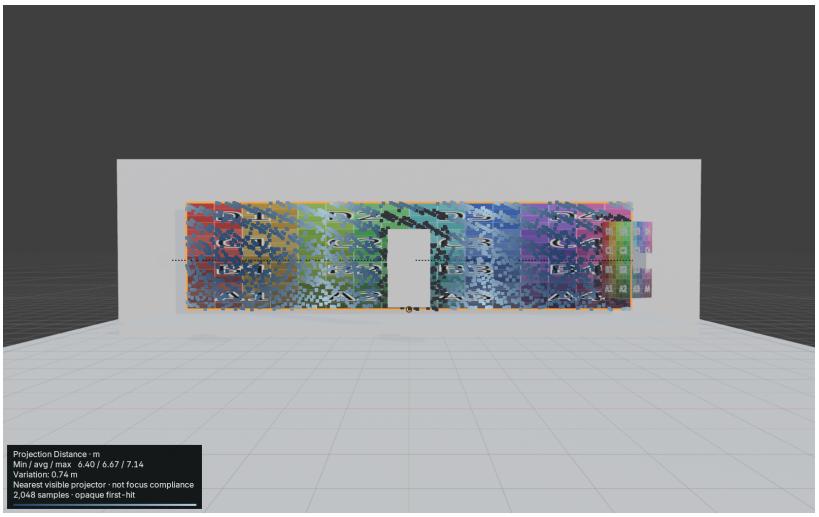


19 Compare Estimated Illuminance

Set Receiver SCR001 and sources Active Blend Group. Move TRAINING_Occluder to X = -9. Choose Illuminance and Analyse. Record min/average/max. Move the blocker to X = 0 and reanalyse; average incident lux should fall. Try Fixed Range with Maximum Lux 1000, then return to Auto. Target Lux is an optional display reference.

Compare projector distances

Use a working copy; restore the projector position before the next exercise.

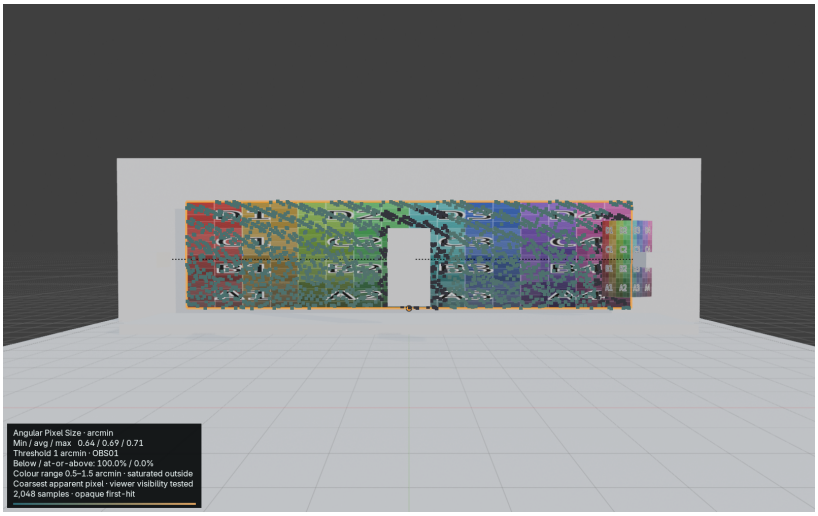


20 Inspect Projection Distance

Choose Distance and Analyse. Record the range and Variation. Select PJ02 and move it one metre farther from the receiver along Y. Reanalyse and compare; nearest-source control may switch locally. Restore its prior position. Inspect TRAINING_Curved_Receiver separately for geometric variation; restore Receiver SCR001 afterwards.

Check pixels from the audience

The original upgraded scene includes OBS01 at Y = -6 m with Eye Height 1.65 m.



21 Move OBS01 and change threshold

Choose Acuity and Observer OBS01. Move its base to Y = -2, Analyse and record angular average. Move to Y = -12 and Analyse; angular pitch should decrease. Return to Y = -6, try threshold 0.7 arcmin and reanalyse. Restore 1 arcminute, then Analyse again. Export the Illuminance, Distance and Acuity views and the complete nine-view Study Package.